Workflow Automation Guide

Configuring DAG Workflows and Analytics Node Library

One Janitorial Operations Library

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1. Overview & Purpose

Technical reference for defining and building Automated Directed Acyclic Graph (DAG) workflows. Explains node connections, parameters, and execution triggers.

Learn how to orchestrate autonomous cleaning campaigns, customer responses, and invoice processing by wiring modular steps together.

Version 3 integrates 13 new Analytics nodes, allowing workflows to ingest files, compute averages, run forecast models, and distribute PDF reports automatically.

2. Key Features & Blueprint

Drag-and-Drop Workflow Canvas: A visual builder allowing designers to drag triggers, conditions, and actions into executable graphs.

Analytics Node Library: Reusable processing blocks (Data Import, Data Cleaner, Aggregation, Forecast, Visualization, Report Generator, Share Link).

Real-Time Execution Inspector: Terminal displaying logs and execution metrics for each active workflow path.

3. System Process Flow

4. Step-by-Step Usage & Operations

Creating a new flow: Click Create Workflow, drag a Trigger node, drag action nodes, connect terminals, and click Validate.

Wiring Analytics Nodes: Connect a CSV Reader node to an Aggregation node, map output values, and feed into a Share Report node.

Deploying to production: Click Deploy to publish workflow. The workflow immediately registers with the execution engine.

5. Best Practices & Safety Guidelines

Avoid circular loops in paths by verifying node dependency directions before deploying.

Ensure every external API call node is followed by a fallback node to handle timeouts.

Always append a Data Cleaner node before Aggregation nodes to filter blank values.

6. Troubleshooting & FAQ

Node validation fails: Inspect connection lines for disconnected inputs or missing parameter values.

Aggregation typecast errors: Ensure the input column datatype is converted to Number before running sum/averages.

Workflow freezes: Check active status of BullMQ worker daemon and Redis database connection.

7. Glossary & Appendix

DAG: Directed Acyclic Graph - A flow model with no circular loops.

Workflow: A sequence of connected nodes representing a business process.

Aggregation: Mathematical summarization of record columns (e.g. sum, count, mean).